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Genetic-Algorithm-Based Camera Placement for Congested Port Traffic: The Port of Algeciras Case Study

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ABSTRACT Optimal placement of cameras in logistics and port environments is a key problem in intelligent surveillance. Most previous work relies on exact optimization formulations, greedy heuristics, or geometric coverage schemes. However, these approaches often struggle with complex vehicle flows or large-scale networks, and they do not easily accommodate operational constraints such as a limited sensor budget. This work proposes a genetic-algorithm-based approach to camera placement for estimating vehicle flows in the Port of Algeciras. Unlike methods that focus on direct visibility or purely spatial coverage, the proposed method operates on the traffic graph and uses flow-reconstruction error metrics as the objective function. The algorithm is evaluated in two scenarios with different topological complexity, one synthetic and one derived from the real port network. The results show that the evolutionary approach can adapt to budget constraints and heterogeneous graph structures, yielding competitive solutions with respect to traditional heuristics reported in the literature. Finally, we discuss modeling limitations and outline future work, including comparisons with classical algorithms, multiobjective formulations, and scenarios with real port traffic data.

INDEX TERMS camera placement, evolutionary optimization, traffic modeling, vehicle flow, smart ports

I. INTRODUCTION

EVERY year, hundreds of thousands of Maghrebi families travel from Europe to North Africa in what is known as the Strait Crossing Operation (OPE). This migratory phenomenon, concentrated in the summer months, generates intense congestion in the main ports of the Iberian Peninsula, especially at the Port of Algeciras—the primary maritime hub connecting Spain and Morocco. During these weeks, the volume of vehicles and passengers can increase exponentially, causing saturation on access highways, national roads, and port areas [1].

Traffic detection and monitoring constitute essential tools to ensure fluidity and safety in this high-demand environment. Efficient management of vehicle flow not only has an economic impact—given the port’s strategic role in international trade and mobility—but also social and national security value, as it reduces waiting times, emissions, and accident risks.

Nevertheless, the problem of traffic monitoring in complex environments such as a port presents multiple challenges. One of the most significant is the optimal placement of

sensors (cameras, radars, or IoT devices) to achieve complete coverage of vehicle flow at a reasonable cost. Inadequate sensor positioning may lead to observational blind spots, redundancies, or the loss of critical information.

This work proposes a formulation of the optimization problem for sensor placement within the Port of Algeciras, considering both the accuracy of traffic estimation and the installation cost. To this end, a genetic algorithm is proposed to minimize estimation errors in traffic conditions while simultaneously optimizing the spatial arrangement and number of sensors.

The main contributions of this work are summarized as follows:

- We formulate the camera placement task as an optimization problem defined over the traffic graph, incorporating flow-based error metrics and operational constraints such as sensor budget.
- We propose a genetic algorithm specifically adapted to the structure of traffic networks, including a fitness function based on flow-reconstruction error and a robust variant inspired by $k + 1$ coverage.

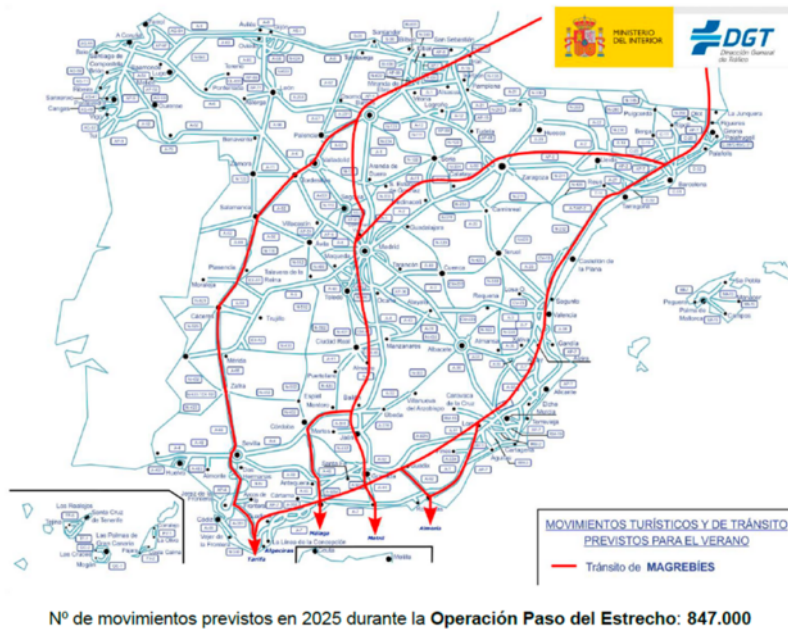


FIGURE 1. Map of Maghrebi traffic from Europe to the main Spanish ports.

- We evaluate the approach in two scenarios with different topological complexity: a synthetic network and a real network derived from the Port of Algeciras.
- We compare the performance of the evolutionary method against simple baselines and discuss its suitability for large-scale and heterogeneous port infrastructures.

The structure of the article is as follows: Section II reviews the most relevant related work. Section III formally introduces the optimal sensor placement problem in traffic networks and outlines its mathematical formulation. Section IV details the methodology, presenting the genetic algorithm and describing each component of the proposed evolutionary approach. Section V presents and analyses the experimental results, including comparison with other test inputs. Finally, Section VI summarizes the main findings and outlines directions for future research.

II. RELATED WORKS

Research on sensor and camera placement in transportation networks has traditionally focused on determining how many sensors are required and where they should be located to maximize information while minimizing redundancy. A well-established line of work models the problem directly on the structure of the traffic network, aiming to identify minimal sensor sets that allow reconstructing the global network state. This approach, explored in depth by [2], provides an algebraic and graphical framework to assess the informativeness of different configurations. From a complementary perspective, several metrics have been proposed to quantify the information gain of candidate layouts. Among them, the

perceptual-entropy metric introduced in [3] enables ranking heterogeneous sensor configurations according to their contribution to state estimation.

To address scalability, submodular optimization has been extensively adopted due to its theoretical approximation guarantees and its ability to handle large networks efficiently, as reviewed in [4] and in the comprehensive survey by [5]. These techniques have been combined with combinatorial heuristics and hybrid optimization methods that attempt to bridge the gap between topological models and their physical implementation. For instance, continuous optimization of camera orientation and position, as presented in [6], adapts deployments to real-world geometry and reduces occlusions—an essential requirement in complex environments such as port access areas. Similarly, [7] proposes an alternating greedy removal-and-placement strategy that seeks to improve coverage while keeping the number of deployed sensors low.

In recent years, artificial intelligence has gained prominence in sensor deployment research. The method proposed in [8] combines deep reinforcement learning with QR-pivoting to select sensor locations that maximize the reconstruction of traffic states, enabling adaptive policies that exploit historical patterns and respond to dynamic changes. In parallel, probabilistic modeling has become a central tool for evaluating and guiding sensor placement decisions. Gaussian-process-based approaches such as [9] identify locations that reduce spatial and temporal uncertainty, providing strong baselines for comparison.

Another important line of research focuses on robustness. The analytical study in [10] examines model-based

approaches that maintain reconstruction quality under sensor failures or partial observations. Complementary proposals integrate sensor selection with data-fusion strategies. For example, [11] demonstrates how combining cameras with additional sensing modalities improves estimation accuracy and provides redundancy during adverse conditions.

Overall, while recent literature emphasizes methods with theoretical guarantees, learning-based strategies, or probabilistic models, metaheuristic optimization remains a highly relevant approach due to its flexibility and low modeling requirements. Genetic algorithms, in particular, can efficiently explore large solution spaces without relying on strong assumptions such as submodularity or differentiability—an advantage in real networks with irregular geometries, operational constraints, and heterogeneous traffic patterns.

In this context, the present work addresses a research gap in flow-based sensor placement methods that operate directly on the traffic graph and incorporate error-driven objectives. Unlike approaches centered on geometric visibility or coverage maximization, our method focuses on reconstructing vehicle flows under practical budget constraints and adapts naturally to the complex topology of port environments such as the Port of Algeciras.



FIGURE 2. Algeciras Port Route Road.

III. PROBLEM STATEMENT

This section defines the representation of the port network, the notation, the measurement model, and the mathematical formulation of the optimal sensor placement problem [2], [4].

The objective is to select a subset of candidate locations that ensures the observability (and its robust counterpart) of internal flows with the lowest possible error under a budget constraint.

A. TIME-EXPANDED NETWORK

The port area will be modeled using a time-dependent directed graph

$$G_{T,\omega} = (\mathcal{N}, \mathcal{A}_{T,\omega}),$$

where each time interval $\tau \in \mathcal{T}$ and operational scenario $\omega \in \Omega$ induces a set of arcs $\mathcal{A}_{\tau, \omega}$ representing:

- (i) movement between checkpoints/cameras,
(ii) storage (carry-over in parking or waiting areas), and
(iii) border (entry/exit points of the area). The subsets of entry and exit arcs are denoted by $\mathcal{E}_\omega$ and $\mathcal{S}_\omega$, respectively.

For a node $i \in \mathcal{N}$, $\delta^+(i)$ and $\delta^-(i)$ denote the set of arcs leaving and entering i . [2], [10]

B. NOTATION, DATA, AND PARAMETERS

- $S \subseteq \bigcup_{\tau, \omega} \mathcal{A}_{\tau, \omega}$: set of candidate arcs for sensor deployment.
- $x_a \in \{0, 1\}$: decision to install a sensor on arc a (1 if installed).
- $c_a \geq 0$: cost of installing a sensor on a ; B is the total budget.
- $f_a^{\tau, \omega} \geq 0$: real flow on arc a in (τ, ω) .
- $\hat{f}_a^{\tau, \omega} \geq 0$: pseudo-measurement/prior (from simulation or aggregated counts) for (a, τ, ω) .
- $\alpha_a \in (0, 1]$: sensor efficiency/detection rate on a (if unknown, it is assumed to be calibrated and embedded into $\hat{f}$).
- $w_a^{\tau, \omega} > 0$: information weight; in practice, a robust choice for L1 is $w_a^{\tau, \omega} \approx 1 / \max(|\hat{f}_a^{\tau, \omega}|, \varepsilon)$, with a small $\varepsilon > 0$ to avoid division by zero.
- $O_i^{\tau, \omega}, D_i^{\tau, \omega}$: observed counts at boundary nodes (zero at interior nodes).
- $\text{cap}_u^{\tau, \omega}$: capacity of storage arcs u (if applicable).
- M : Big-M constant operationally bounded (e.g., maximum flow per interval estimated from the highest observed inflow at entry arcs). [8], [9]
- $z_i^{\tau, \omega} \geq 0$: slack variable for flow-balance violations at node i .
- $\lambda > 0$: penalty parameter associated with flow-balance slacks in the objective.
- $\varepsilon > 0$: small regularization constant used in the definition of $w_a^{\tau, \omega}$.

C. MEASUREMENT MODEL

If a sensor is installed on arc a , the observed value is modeled as

$$y_a^{\tau,\omega} = \alpha_a f_a^{\tau,\omega} + \epsilon_a^{\tau,\omega}, \quad \mathbb{E}[\epsilon_a^{\tau,\omega}] = 0, \quad \text{Var}(\epsilon_a^{\tau,\omega}) = \sigma_a^2. \quad (1)$$

where the systematic component α_a accounts for the efficiency of the device. In arcs without a sensor ($x_a = 0$), the flow is inferred through balance and topological constraints. [4], [10]

D. FLOW CONSERVATION AND CAPACITIES

For each node $i \in \mathcal{N}$ and for every (τ, ω) , flow conservation is enforced as:

$$\sum_{a \in \delta^+(i)} f_a^{\tau, \omega} - \sum_{a \in \delta^-(i)} f_a^{\tau, \omega} = O_i^{\tau, \omega} - D_i^{\tau, \omega}, \quad (2)$$

where $O_i^{\tau,\omega}$ and $D_i^{\tau,\omega}$ are nonzero only at boundary nodes. For storage arcs u , capacity bounds are imposed as:

$$0 \leq f_y^{\tau, \omega} \leq \text{cap}_y^{\tau, \omega}. \quad (3)$$

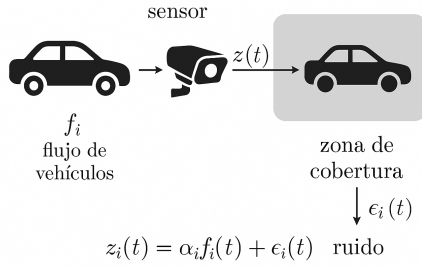


FIGURE 3. Vehicle detection flow.

In practice, when working with discretized flows obtained from microscopic simulation, exact conservation at every (i, τ, ω) may be incompatible with measurement noise and aggregation. In the implementation used in this work, (2) is therefore relaxed by introducing nonnegative slack variables $z_i^{\tau, \omega}$ and adding a penalty term $\lambda \sum_{\tau, \omega} \sum_{i \in \mathcal{N}} z_i^{\tau, \omega}$ to the objective, thus encouraging approximate conservation while allowing small imbalances.

E. OBSERVABILITY AND ROBUSTNESS

To ensure the identifiability of internal flows without estimating O/D matrices, coverage of the graph's cyclic structure and relevant cuts is required. Let $\mathcal{C}_{\tau, \omega}$ denote a directed cycle basis (e.g., obtained via Paton's algorithm), and let $\mathcal{K}_{\tau, \omega}$ be a set of relevant minimum cuts (boundary–interior separators). The robust version, accounting for the possible loss of up to k sensors, is expressed through $(k+1)$ -coverage:

$$\sum_{a \in C} x_a \geq k+1, \quad \forall C \in \mathcal{C}_{\tau, \omega}, \quad (4)$$

$$\sum_{a \in K} x_a \geq k+1, \quad \forall K \in \mathcal{K}_{\tau, \omega}. \quad (5)$$

This formulation avoids combinatorial enumeration of failures, maintaining scalability across discretized time horizons.

F. PROBLEM FORMULATION (MILP)

A weighted L1-norm objective function (robust to outliers) will be used, with residuals activated only when a sensor is present. Variables $r_a^{\tau, \omega} \geq 0$ will be introduced to quantify the discrepancy between the prediction (via (1) with $z \approx \hat{f}$) and

the pseudo-measurement $\hat{f}_a^{\tau, \omega}$:

$$\min_{x, f, r} \sum_{\tau \in \mathcal{T}} \sum_{\omega \in \Omega} \sum_{a \in \mathcal{A}_{\tau, \omega}} w_a^{\tau, \omega} r_a^{\tau, \omega} \quad (6)$$

s.a. Flow conservation and capacities (2)–(3),

$$\sum_{a \in \mathcal{S}} c_a x_a \leq B, \quad (7)$$

Coverage/robustness (4)–(5),

$$r_a^{\tau, \omega} \geq \alpha_a f_a^{\tau, \omega} - \hat{f}_a^{\tau, \omega} - M(1 - x_a), \quad (8)$$

$$r_a^{\tau, \omega} \geq \hat{f}_a^{\tau, \omega} - \alpha_a f_a^{\tau, \omega} - M(1 - x_a), \quad (9)$$

$$x_a \in \{0, 1\}, \quad f_a^{\tau, \omega} \geq 0, \quad r_a^{\tau, \omega} \geq 0. \quad (10)$$

In the computational experiments, the weights $w_a^{\tau, \omega}$ are chosen according to $w_a^{\tau, \omega} \approx 1/\max(|\hat{f}_a^{\tau, \omega}|, \epsilon)$, providing an approximate normalization by the magnitude of the pseudo-measurements and mitigating the tendency to concentrate on low-flow arcs. Moreover, the relaxed flow-balance formulation described in Section III-D is used, incorporating the slack variables $z_i^{\tau, \omega}$ and their penalty into the objective.

G. NOTATION

TABLE 1. Brief List of Notation.

Symbol	Description
$\mathcal{N}, \mathcal{A}_{\tau, \omega}$	Nodes and arcs of the time-expanded graph
T, Ω	Time intervals and operational scenarios
$\mathcal{E}_{\omega}, \mathcal{S}_{\omega}$	Inflow and outflow arcs (boundary)
$\mathcal{S}$	Set of candidate arcs for sensor placement
x_a	Binary installation variable on arc a
c_a, B	Unit cost and total budget
$f_a^{\tau, \omega}, \hat{f}_a^{\tau, \omega}$	Actual flow / pseudo-measurement at (a, τ, ω)
$y_a^{\tau, \omega}$	Sensor measurement at (a, τ, ω)
α_a, σ_a	Sensor efficiency and noise deviation
$w_a^{\tau, \omega}$	Information weight (e.g. $1/\max(\hat{f}_a^{\tau, \omega} , \epsilon)$)
$O_i^{\tau, \omega}, D_i^{\tau, \omega}$	Counts at boundary nodes (in/out)
$\text{cap}_u^{\tau, \omega}$	Capacity on storage arcs u
$\mathcal{C}_{\tau, \omega}, \mathcal{K}_{\tau, \omega}$	Cycle basis and relevant cuts
M	Operational Big-M bound per interval
$r_a^{\tau, \omega}$	Absolute residual at (a, τ, ω)
$z_i^{\tau, \omega}$	Flow-balance slack at node i
λ	Penalty parameter for flow-balance slacks
ϵ	Regularization constant in the weight definition

IV. EVOLUTIONARY COMPUTATIONAL APPROACH

This section describes the design of the genetic algorithm (GA) used to solve the optimal sensor placement problem defined in ???. The GA operates on binary decisions $x \in \{0, 1\}^{|\mathcal{S}|}$ and, to evaluate each individual, fixes x and solves (exactly or approximately) the continuous problem in f, r , consistent with (2)–(3) and (8)–(9). The approach avoids enumerating subsets of failures and uses the $(k+1)$ -coverage from (4)–(5) to ensure robustness.

A. REPRESENTATION AND FEASIBILITY

Each individual is a binary vector $x = (x_a)_{a \in \mathcal{S}}$. Feasibility with respect to budget and coverage is ensured through a repair operator:

- **Budget** ((7)): if $\sum_a c_a x_a > B$, sensors are removed until the budget is satisfied, starting with those having the lowest benefit-to-cost ratio.
- **Robust coverage** ((4)–(5)): if any $C \in \mathcal{C}_{\tau, \omega}$ or $K \in \mathcal{K}_{\tau, \omega}$ does not meet the $(k+1)$ -coverage condition, sensors are added by prioritizing arcs with the highest frequency of appearance in cycles/cuts per cost.

The benefit-to-cost metric for the repair operator can be defined as

$$s_a = \frac{\gamma_1 \text{freq}(a; \mathcal{C} \cup \mathcal{K}) + \gamma_2 \bar{w}_a}{c_a},$$

where freq counts how many cycles/cuts arc a appears in, and $\bar{w}_a$ is the average information weight of a over the time horizon. Fixed coefficients $\gamma_1, \gamma_2 > 0$ are used (e.g., $\gamma_1=1, \gamma_2=1$).

B. EVALUATION (“INNER SOLVE”)

For a given x , two evaluation modes are considered:

a: Exact (LP).

The linear problem in f, r is solved with x fixed, using (2)–(3), (8)–(9), and an operationally bounded M . This yields $r_a^{\tau, \omega}(x)$ and a direct objective value consistent with (6).

b: Surrogate (fast).

A fast projection of pseudo-measurements $\hat{f}$ is used to obtain a feasible $\tilde{f}$ (e.g., iterative nodal balancing over 5–10 iterations with truncation to capacity limits). Then, for instrumented arcs ($x_a=1$), the approximate residual is computed as $r_a^{\tau, \omega}(x) \approx |\alpha_a \tilde{f}_a^{\tau, \omega} - \hat{f}_a^{\tau, \omega}|$; if $x_a=0$, the residual is not computed (ignored in the fitness).

c: Revalidation.

To prevent surrogate drift, elite individuals (top 5–10% of the population) and the best new individuals every G_{val} generations (e.g., every 5–10) are re-evaluated using the exact mode.

C. FITNESS FUNCTION (WEIGHTED L1 NORM)

To maintain consistency with III-F, a weighted and normalized L1 metric is used:

$$\begin{aligned} \text{Weighted } R(x) &= \sum_{\tau, \omega} \sum_{a \in \mathcal{A}_{\tau, \omega}} w_a^{\tau, \omega} r_a^{\tau, \omega}(x) \mathbb{I}\{x_a = 1\} \\ \text{Normalized } N(x) &= \frac{1}{|\mathcal{T}| |\Omega|} \cdot \frac{1}{\max\{1, \sum_a x_a\}} \\ \text{Combined } \bar{r}(x) &= N(x) R(x) \end{aligned} \quad (11)$$

The **fitness** to be minimized is:

$$\text{fit}(x) = \alpha \bar{r}(x), \quad (12)$$

with $\alpha > 0$ (by default, $\alpha = 1$). Since the budget is already a hard constraint and robustness is ensured by the repair operator, no additional penalty terms are included in (12) to avoid double counting.

D. INITIALIZATION AND OPERATORS

a: Initialization (three seeds).

- (i) Random under budget,
- (ii) Greedy, adding sensors that maximize s_a (coverage from (4)–(5) up to $(k+1)$ -coverage),
- (iii) Robust, prioritizing arcs with high freq in $\mathcal{C}$ and $\mathcal{K}$.

b: Selection.

Tournament of size 2–3.

c: Crossover.

Uniform, with probability $p_c \in [0.7, 0.9]$, preserving blocks that cover cycles/cuts (if recombination violates coverage or budget, the repair operator is applied).

d: Mutation.

Bit-flip with rate

$$p_m \in \left[\frac{1}{|\mathcal{S}|}, \min\{0.1, \frac{3}{|\mathcal{S}|}\} \right],$$

Bit-flip with a bias toward activating arcs that increase coverage where deficits exist.

e: Elitism.

Preservation of the 2–4 best individuals per generation.

E. HYPERPARAMETERS AND STOPPING CRITERION

Recommended reproducible configuration:

$$N_{\text{pop}} = 50\text{--}100, \quad G_{\text{max}} = 200\text{--}300, \quad G_{\text{val}} = 5\text{--}10.$$

Stopping criterion based on stagnation: if the best fit does not improve for $G_{\text{stop}} = 30$ consecutive generations, stop.

F. COMPACT PSEUDOCODE

Input: graph $G_{\tau, \omega}$, sets $\mathcal{S}, \mathcal{C}, \mathcal{K}$, budget B , parameters α (and optionally β), k , bounded M , weights w . **Output:** vector x^* .

- 1: Generate initial population using the three seeds and fill randomly
- 2: Apply repair operator
- 3: **for** $g = 1 \dots G_{\text{max}}$ **do**
- 4: Evaluate individuals using the surrogate
- 5: **if** $g \bmod G_{\text{val}} = 0$ **then**
- 6: Reevaluate elites with the exact LP
- 7: **end if**
- 8: Select via tournament
- 9: Apply uniform crossover with probability p_c
- 10: Apply mutation with probability p_m
- 11: Repair offspring (budget and $(k+1)$ -coverage)
- 12: Perform generational replacement with elitism
- 13: Update best solution found
- 14: **if** stagnation $\geq G_{\text{stop}}$ **then**
- 15: **break**
- 16: **end if**
- 17: **end for**
- 18: Return best individual x^* (reevaluated exactly)

G. BASELINES AND LIGHTWEIGHT VALIDATION

To interpret results, it is recommended to compare (conceptually or in lightweight experiments) with:

- (i) Greedy submodular coverage (cycles/cuts) under budget,
- (ii) Selection by QR-pivoting followed by flow reconstruction,
- (iii) Probabilistic benchmark (e.g., Gaussian process) to validate the uncertainty reduction.

These references situate the GA in terms of cost–benefit and reconstruction quality.

H. PRACTICAL CONSIDERATIONS

- (i) **Consistency with IV:** use the same L1 norm and the same weights $w_a^{\tau, \omega}$ as in (6).
- (ii) **Big-M:** set M as an operational bound (maximum in-flow per interval) to avoid weak relaxations when evaluating the exact LP.
- (iii) **Evaluation cache:** memoize $\text{fit}(x)$ for repeated individuals; reuse coverage factors (frequencies of a in $\mathcal{C}, \mathcal{K}$).
- (iv) **Scalability:** the exact phase acts only on elites; the rest uses a surrogate, keeping times bounded.

V. RESULTS

This section presents the results obtained in the different stages of the work:

- 1) the generation and simulation of traffic in SUMO;
- 2) the heuristic optimization of camera locations by means of a Genetic Algorithm (GA);
- 3) the experimental design for the MILP model of optimal sensor placement.

A. EXPERIMENTAL SETUP IN SUMO

To evaluate the proposed sensor placement strategies, two simulation scenarios were designed in *Simulation of Urban Mobility* (SUMO). These scenarios are later used as a common testbed for both the GA and the MILP model:

- **Scenario A (normal operation):** port network with pseudo-randomly generated routes, allowing us to test the behavior of the algorithm without operational restrictions. It is used to validate the robustness of the approach in a controlled setting.
- **Scenario B (OPE port routes):** this scenario incorporates the routes associated with the embarkation of vehicles bound for Tanger Med during the “Operación Paso del Estrecho” (OPE). It corresponds to Phase II of the real port operation, in which additional parking areas (Llano Amarillo) are opened. Accurately estimating the actual traffic flow and parking occupancy in these plots is critical, since excessive saturation triggers the activation of additional parking areas in the fishing port. This may lead to a ticket interchangeability event, where each vehicle can embark with any shipping company (analogous to an airport where Ryanair passengers could fly with Iberia if seats are available).

In both scenarios, synthetic *routes* are generated from hourly demand distributions, and multiple simulations with different random seeds are run in order to reduce sampling variability.

B. EXPERIMENTAL SETUP OF THE GENETIC ALGORITHM

The proposed GA was evaluated on the two SUMO-based scenarios described above for the camera placement problem in the Port of Algeciras:

- **Scenario A (randomized flows):** routes are generated so that vehicles travel between randomly chosen origins and destinations within the port area. This approximates vehicle flows during non-OPE periods.
- **Scenario B (port routes):** this scenario is built from the real port map. Vehicle flows, although synthetic, are more realistic and reflect typical port traffic patterns, especially at roundabouts and access ramps. The network in this scenario contains 93 edges.

The main GA parameters are summarized in Table 2. The implementation relies on the DEAP framework and uses a minimization fitness, where each individual encodes a binary sensor placement and is evaluated through the domain-specific function `evaluate_sensor_placement` under a fixed sensor budget. :contentReference[oaicite:1]index=1

TABLE 2. Genetic Algorithm simulation parameters

Parameter	Value
Population size μ	50
Offspring size λ	50
Generations	50
Selection	Tournament (size = 3)
Crossover probability	80%
Mutation probability	20%

Each scenario was executed multiple times until stable solutions close to the best observed values were obtained.

C. EVALUATION METRICS

The following metrics are used in the evaluation:

- **RMSE:** root mean squared error between the true flow (from SUMO) and the flow estimated from the selected camera configuration.
- **Fitness:** objective value returned by the domain-specific evaluation function. In our implementation, this fitness corresponds to the RMSE (lower is better), possibly including internal penalties when the sensor budget is exceeded. For clarity, we report it explicitly as RMSE.
- **Number of sensors:** the algorithm must select sensors under a fixed budget. In Scenario A the budget is set to 100 sensors, whereas in Scenario B the total budget is 50 sensors.

All metrics are computed from SUMO outputs (flows and travel times) for each candidate sensor configuration proposed by the GA.

D. MAIN GA RESULTS

Table 3 reports the best solutions found by the GA in both scenarios.

TABLE 3. Best solutions found by the GA

Scenario	Fitness (RMSE)	Sensors	Time (s)	RMSE
A	102.8969	98	821.32	2.8969
B	4.6805	50	8.27	4.6805

In Scenario A (randomized flows), the best fitness value corresponds to an RMSE of 2.8969. In Scenario B (more realistic flows), the best RMSE obtained was 4.6805. Since the fitness function in the current GA implementation is aligned with the RMSE, it essentially captures the reconstruction error associated with each sensor configuration under the specified budget.

Figures 4 and 6 show the evolution of RMSE across generations for Scenarios A and B, respectively, while Figures 5 and 7 illustrate the spatial distribution of sensors in the best solutions identified by the GA.

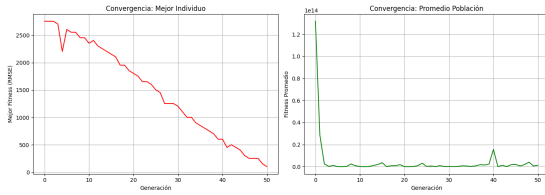


FIGURE 4. RMSE across generations in Scenario A.



FIGURE 5. Sensor distribution for the best GA solution in Scenario A.

E. MILP EXPERIMENTAL DESIGN

In addition to the heuristic GA approach, a Mixed-Integer Linear Programming (MILP) model is formulated for the

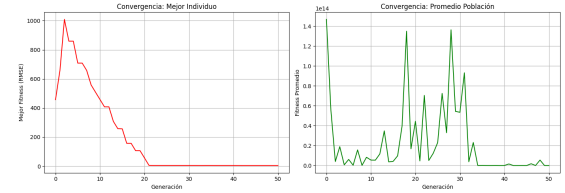


FIGURE 6. RMSE across generations in Scenario B.



FIGURE 7. Sensor distribution for the best GA solution in Scenario B.

optimal sensor placement problem on the port network. The main objectives of the MILP are to:

- ensure observability of internal flows without explicitly estimating a full origin–destination matrix;
- minimize an aggregated measure of reconstruction error (e.g., a weighted L_1 norm of the discrepancies between pseudo-measurements and reconstructed flows) under a budget constraint;
- incorporate robustness constraints against the loss of up to k sensors via $(k+1)$ -coverage conditions on relevant cycles and cuts of the network.

The experiments planned with the MILP follow the problem formulation introduced in Section III. Both scenarios are considered jointly, while explicitly marking as critical those links that are involved in the OPE operations, in particular the embarkation flows of vehicles bound for Tanger Med.

Figure 8 shows an example solution of the MILP model with a budget of 20 sensors, where the solver places devices in strategically selected edges of the port network.

In this first version of the work, the experimental design is fully specified: both approaches (GA and MILP) will be evaluated on the same SUMO networks and time horizons, using the same pseudo-measurements (flows and counts on entry and exit arcs) and the same budget.



FIGURE 8. Example distribution of 20 sensors obtained with the MILP model.

F. COMPARISON BETWEEN GA AND MILP

For larger sensor budgets, the current MILP implementation becomes numerically challenging and does not yield solutions within reasonable time. Therefore, we focus the comparison on a budget of $B = 20$ sensors in Scenario B.

Figure 9 compares the flow coverage achieved by the GA and the MILP solution for this setting.

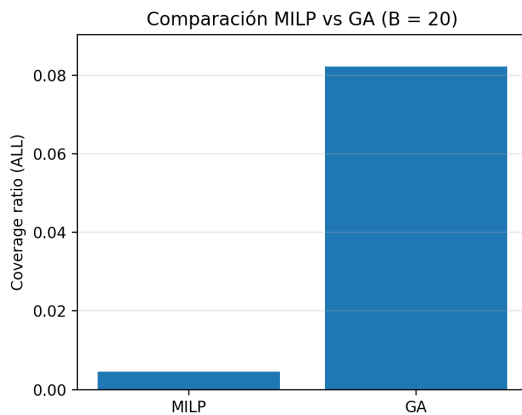


FIGURE 9. Flow coverage comparison between MILP and GA (Scenario B, $B = 20$).

The GA tends to achieve higher overall traffic coverage, as it directly optimizes the reconstruction error over the simulated flows. However, the sensors selected by the MILP are placed precisely where they are most needed to reconstruct the flow structure, and only once these structural constraints are satisfied does the model allocate sensors to other areas of the port. In particular, the MILP solution is able to place sensors in each of the key parking plots involved in the OPE operation.

Figure 10 shows the spatial configuration obtained by the GA for the same budget and scenario, which is less spread across the network in this scenario.

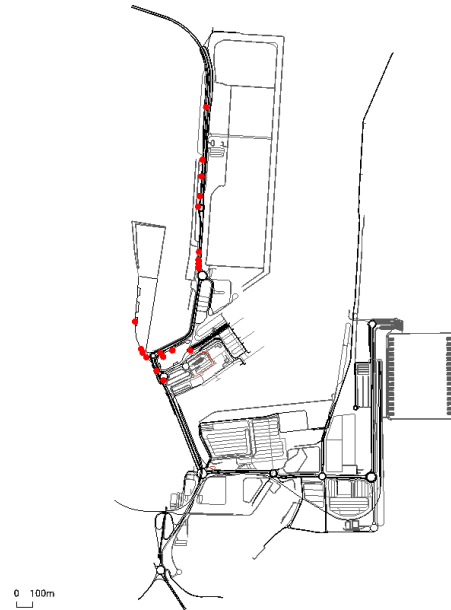


FIGURE 10. Distribution of 20 sensors selected by the GA in Scenario B.

VI. CONCLUSION

A. DISCUSSION

The results show that the Genetic Algorithm can generate high-quality solutions for the camera placement problem under a fixed sensor budget. In Scenario A, the RMSE obtained is significantly lower than in Scenario B, which is consistent with the structure of each map: the random scenario exhibits a more uniform distribution of edges, whereas the realistic port scenario contains critical areas where traffic is highly concentrated and unevenly distributed.

The difference observed in execution times is also expected. Scenario A contains more than four times as many edges as Scenario B, so the computational cost associated with fitness evaluation grows approximately with the size of the graph. This higher complexity explains the considerably longer time required to complete the runs in Scenario A.

The sensor budget is another key factor. In Scenario A, the maximum number of sensors is 100, which allows a larger portion of the network to be monitored and leads to lower final error. In contrast, Scenario B is limited to 50 sensors, which constrains the algorithm's ability to cover certain critical areas of the port graph.

The preliminary comparison with the MILP model suggests a complementary behavior between both approaches. While the GA is effective at reducing the reconstruction error on simulated flows, the MILP tends to concentrate sensors in structurally critical edges (such as parking plots and key bifurcations), which is desirable when observability guarantees are required.

Finally, it is important to note that the fitness used in the GA is a simplification of the real problem. Although it captures the general idea of coverage and reconstruction error, it does not incorporate additional factors such as camera field of view, occlusions, viewing direction or obstacles. Therefore, although the results are consistent and useful for studying global placement strategies, they should be interpreted with caution when extrapolating to real deployments.

B. FUTURE WORK

This work opens up several directions for future research that would bring the study closer to an operational deployment and improve the scientific validity of the results. The main recommended actions are:

- **Validation with real port data.** Integrate real flow and event data provided by the Port Authority to validate the effectiveness of the solutions under actual operating conditions.
- **Heterogeneities in the network.** At present, the different service times of passenger activities (e.g., passport control) are not modeled explicitly. This implies that placing sensors in different parts of the same edge is treated as equivalent. Future work could model these effects as vehicle dwell times in specific segments, assigning higher weights to points with higher variance.
- **Systematic comparisons with baselines.** Evaluate the GA against reference methods such as greedy heuristics, random search, Simulated Annealing (SA), Particle Swarm Optimization (PSO) and, when feasible, exact ILP formulations on small instances. Report means, standard deviations and statistical tests (Shapiro–Wilk, paired *t*-test or Wilcoxon).
- **More realistic camera model.** Extend the objective function to include camera field of view, occlusions, viewing direction, angle of view and installation cost so as to obtain operationally viable placement patterns.
- **Vehicle-type differentiation.** Incorporate traffic classes (cars, trucks, trailers, special vehicles) in the flow model to prioritize surveillance along critical routes according to vehicle type.
- **Multi-objective optimization.** Reformulate the problem as a multi-objective one (e.g., jointly minimizing RMSE and cost), using multi-objective algorithms (NSGA-II, SPEA2) and visualizing the Pareto front.
- **Sensitivity and ablation studies.** Analyze sensitivity to hyperparameters (population size, mutation, crossover), to the budget and to variations in flow generation; perform ablation studies to quantify the contribution of each operator.
- **Dynamic scenarios and temporal patterns.** Model temporal variations (peak hours, ship schedules) and evaluate adaptive strategies or online re-planning.
- **Acceleration and scalability.** Parallelize fitness evaluation, apply surrogate modelling and caching techniques, and explore hybrid approaches (GA + local search) to reduce computation times.

- **Field trials and operational assessment.** Conduct controlled pilots in specific port zones to evaluate technical and operational feasibility, including qualitative feedback from operators.
- **Ethical and privacy aspects.** Study the implications in terms of privacy, data protection and regulatory compliance when deploying cameras in real environments.

Reproducibility. To facilitate reproducibility, the complete source code of the project is provided in an open repository [12].

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Any remaining errors are the sole responsibility of the authors.

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